

ROTARYLINK

Type:

Axis Command.

Syntax:

```
ROTARYLINK(distance, link_dist, acc, sync, link_axis [,  
link_options[, sync_pos]])
```

Alternate Format:

RL ()

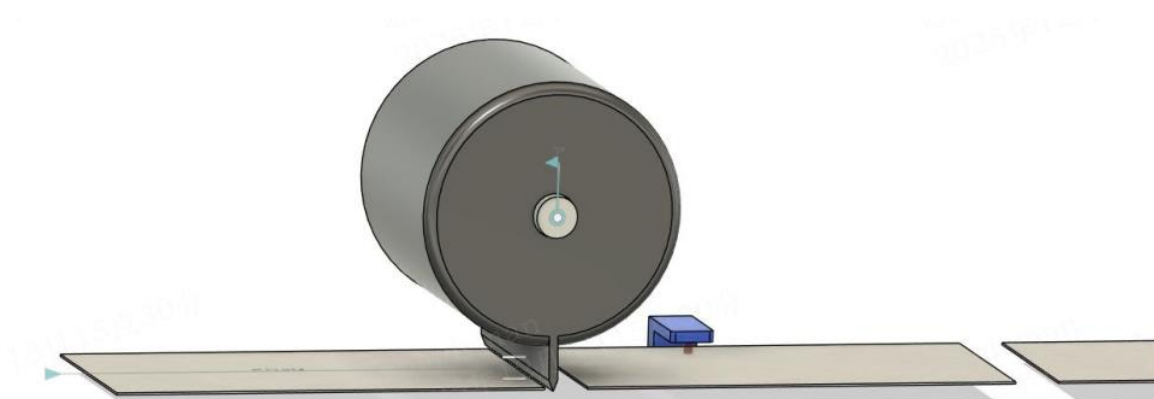
Description:

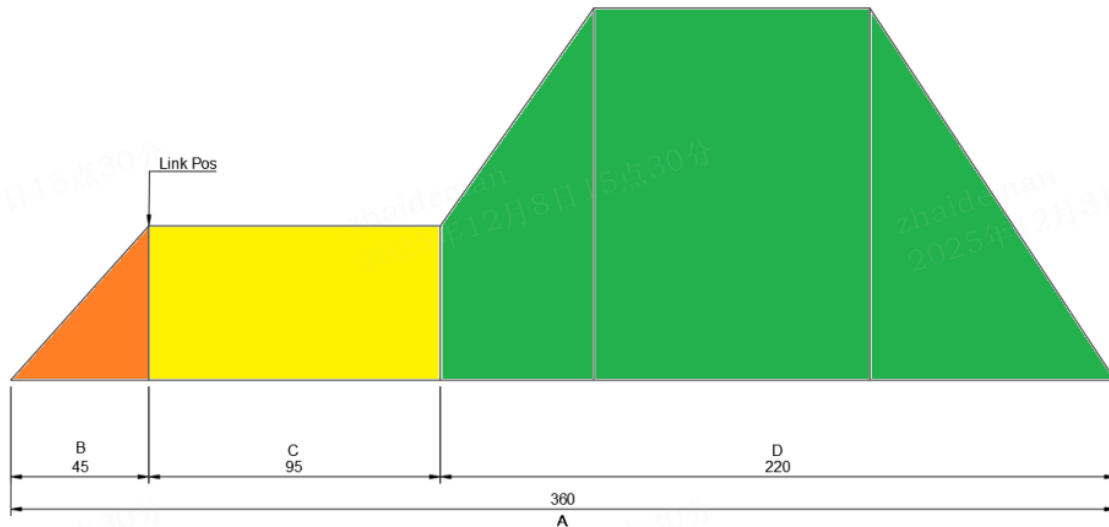
The linked move command is designed for controlling movements such as:

- Rotary knife
- Printing drums
- Label applicators
- Die Cutters

The motion consists of a linear movement with separately variable acceleration and deceleration phases linked via a software gearbox to the MEASURED position (MPOS) of another axis. The command uses the BASE and AXIS, and unit conversion factors in a similar way to other move commands.

Unlike MOVELINK the phase distances are for the base axis, not the link axis.





Parameters:

distance: (A)	incremental distance in user units to be moved on the current base axis, as a result of the measured movement on the linked axis	
Link_dist:	positive incremental distance in user units which is required to be measured on the “link” axis to result in the motion on the base axis.	
acc: (B)	incremental distance in user units to be moved on the current base axis during the acceleration phase	
sync: (C)	incremental distance in user units to be moved on the current base axis during the synchronised phase	
link axis:	Specifies the axis to “link” to. It should be set to a value between 0 and the number of available axes.	
link_options:	Bit value options to customise how your ROTARYLINK operates	
	Bit 0	Choose the start mode.
	Bit 1	0 - link commences at an absolute position on link axis (see sync_pos for start position) 1 - Link commences exactly when registration event MARK occurs on link axis 2 - Link commences exactly when registration event MARKB occurs on link axis 3 - Link commences exactly when registration event

		R_MARK occurs on link axis. (see sync_pos for channel number)
	Bit 2	Choose acceleration and deceleration profile.
	Bit 3	0 - Trapezoidal profile
	Bit 4	1 - Sine speed Profile
		2 - Power 9 polynomial speed
		3 - Power 7 polynomial speed
		4 - Power 5 polynomial speed
	Bit 5	If this bit is set the ROTARYLINK will merge 2 consecutive ROTARYLINK commands: phase D and B will be merged to achieve the Sync_pos without stopping.
	Bit 6	Slave axis follows master axis DPOS
sync_pos:	link_option 0 - the absolute position on the link axis in user UNITS where the Sync phase should start. link_option 1 – the registration channel to start the movement on.	

*When the sync position is set, the sync position of the link axis should be greater than acceleration distance of link axis (link_acc , which can be calculate by acceleration distance of base distance: $link_acc = acc / distance * link_dist$). If not, an exception will be thrown, parameter 7 out of range.*

If you merge 2 consecutive ROTARYLINK, the sync_pos of the second command should greater than the end position of sync phase of the first command.

Examples:

Example 1:

The ROTARYLINK movement is performed under the absence parameter, and the base axis starts to move directly when the link axis moves. The oscilloscope shows that the base axis accelerated for 10, and then moved at a constant speed of 60.

' Data Initialize

```

FOR i = 0 TO 1
  BASE(i)
  UNITS = 100000
  SPEED = 30
  ACCEL = 100
  DECEL = 100
  JERK = 100

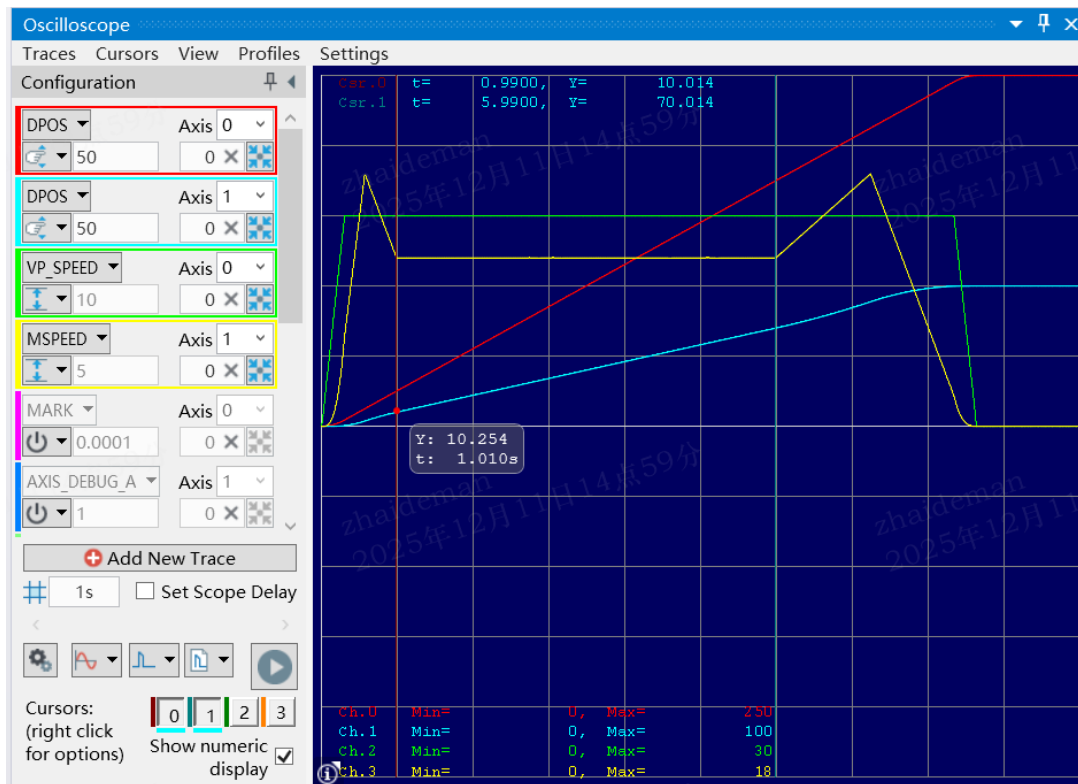
```

DEFPOS(0)
NEXT i

base_dist = 100 ' incremental distance of base axis
link_dist = 250 ' incremental distance of link axis

base_acc = 10 ' incremental distance of the base axis in the acceleration phase
base_sync = 60 ' incremental distance of the base axis in the sync phase
link_axisnum = 0 ' link axis number

BASE(1) ' base axis - 1 , link axis - 0
ROTARYLINK(base_dist, link_dist, base_acc, base_sync, link_axisnum)
MOVE(250) AXIS(0)



Example 2:

Use parameter to set the absolute position on the link axis in user UNITS where the Sync phase should start. In this case, when the link axis moves to the 60, the base axis moves into synchronization phase.

' Data Initialize

```

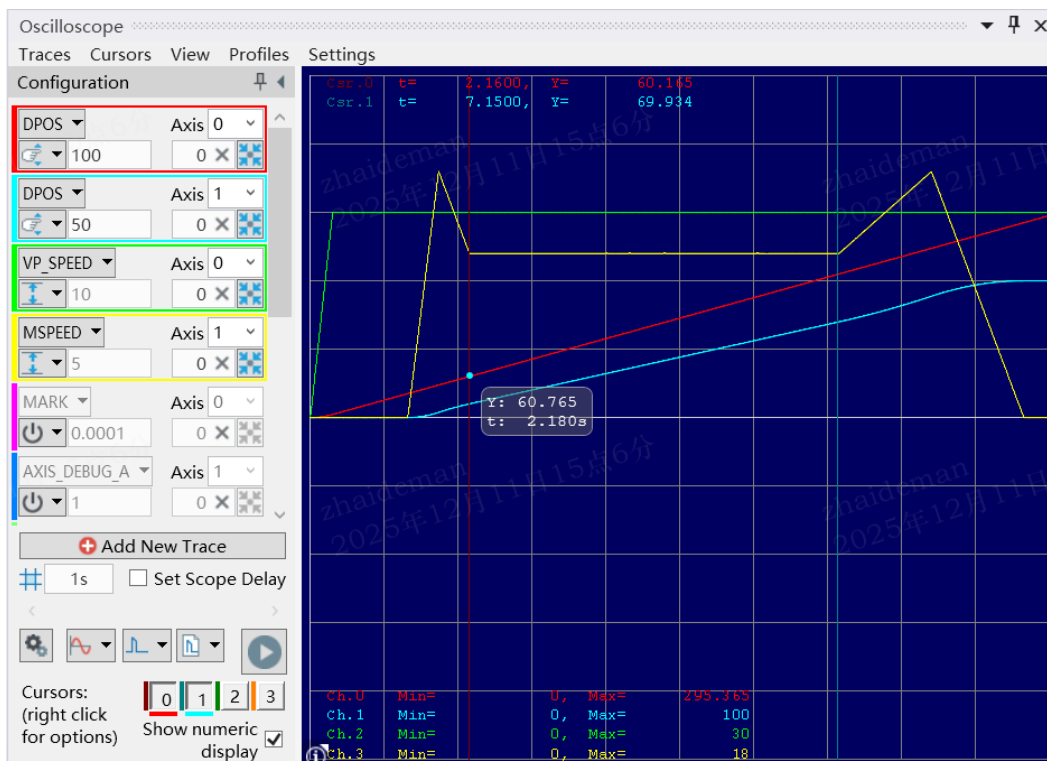
FOR i = 0 TO 1
  BASE(i)
  UNITS = 100000
  SPEED = 30
  ACCEL = 100
  DECEL = 100
  JERK = 100
  DEFPOS(0)
NEXT i

```

base_dist = 100 ' incremental distance of base axis
link_dist = 250 ' incremental distance of link axis

base_acc = 10 ' incremental distance of the base axis in the acceleration phase
base_sync = 60 ' incremental distance of the base axis in the sync phase
link_axisnum = 0 ' link axis number
link_options = 0 ' link commences at an absolute position on link axis
sync_pos = 60 ' where the Sync phase should start on link axis

BASE(1) ' base axis - 1 , link axis - 0
ROTARYLINK(base_dist, link_dist, base_acc, base_sync, link_axisnum, _
link_options, sync_pos)
MOVE(310) AXIS(0)



Example 3:

Choose sine speed profile to be the acceleration and deceleration profile and set the sync phase starts when link axis moves to 60.

' Data Initialize

```
FOR i = 0 TO 1
  BASE(i)
  UNITS = 100000
  SPEED = 30
  ACCEL = 100
  DECEL = 100
  JERK = 100
  DEFPOS(0)
NEXT i
```

base_dist = 100 ' incremental distance of base axis

link_dist = 250 ' incremental distance of link axis

base_acc = 10 ' incremental distance of the base axis in the acceleration phase

base_sync = 60 ' incremental distance of the base axis in the sync phase

link_axisnum = 0 ' link axis number

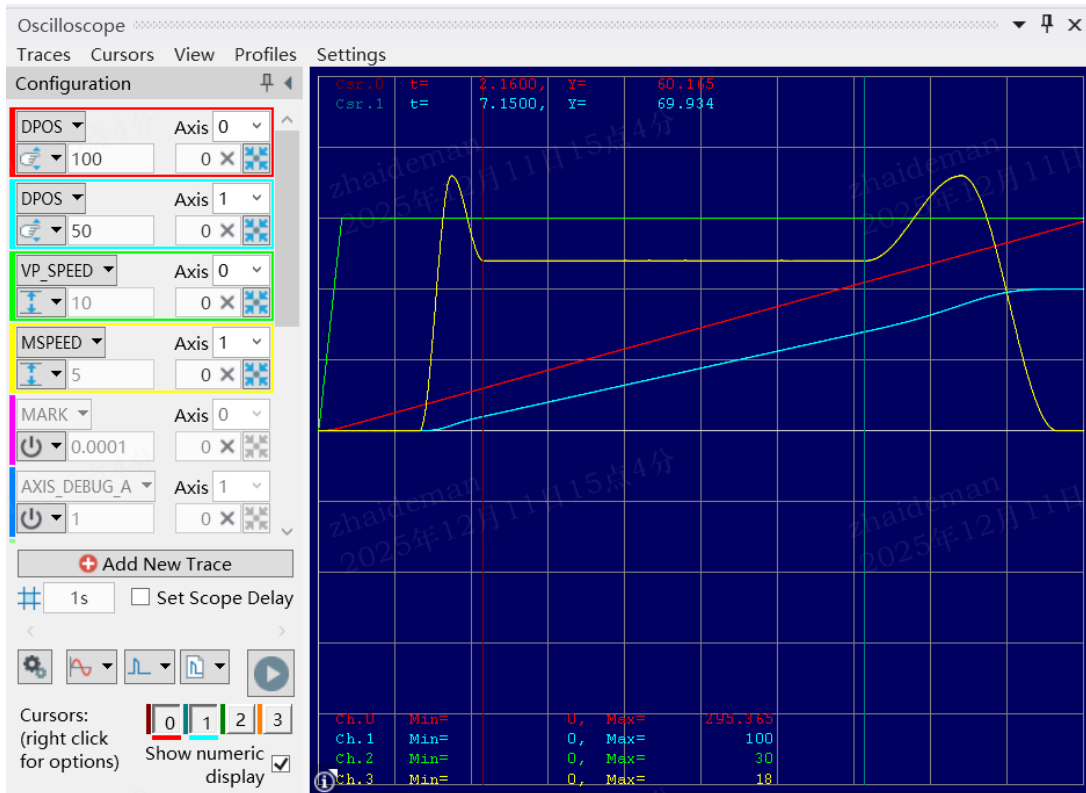
link_options = 4 ' choose acceleration and deceleration profile, 4 - Sine speed Profile

sync_pos = 60 ' where the Sync phase should start on link axis

BASE(1) ' base axis - 1 , link axis - 0

ROTARYLINK(base_dist, link_dist, base_acc, base_sync, link_axisnum, _
link_options, sync_pos)

MOVE(310) AXIS(0)



Example 4:

Merge two ROTARYLINK commands. The second movement start sync phase when link axis at 270.

' Data Initialize

```
FOR i = 0 TO 1
  BASE(i)
  UNITS = 100000
  SPEED = 30
  ACCEL = 100
  DECEL = 100
  JERK = 100
  DEFPOS(0)
NEXT i
```

base_dist = 100 ' incremental distance of base axis

link_dist = 250 ' incremental distance of link axis

base_acc = 10 ' incremental distance of the base axis in the acceleration phase

base_sync_1 = 60 ' incremental distance of the base axis in the sync phase

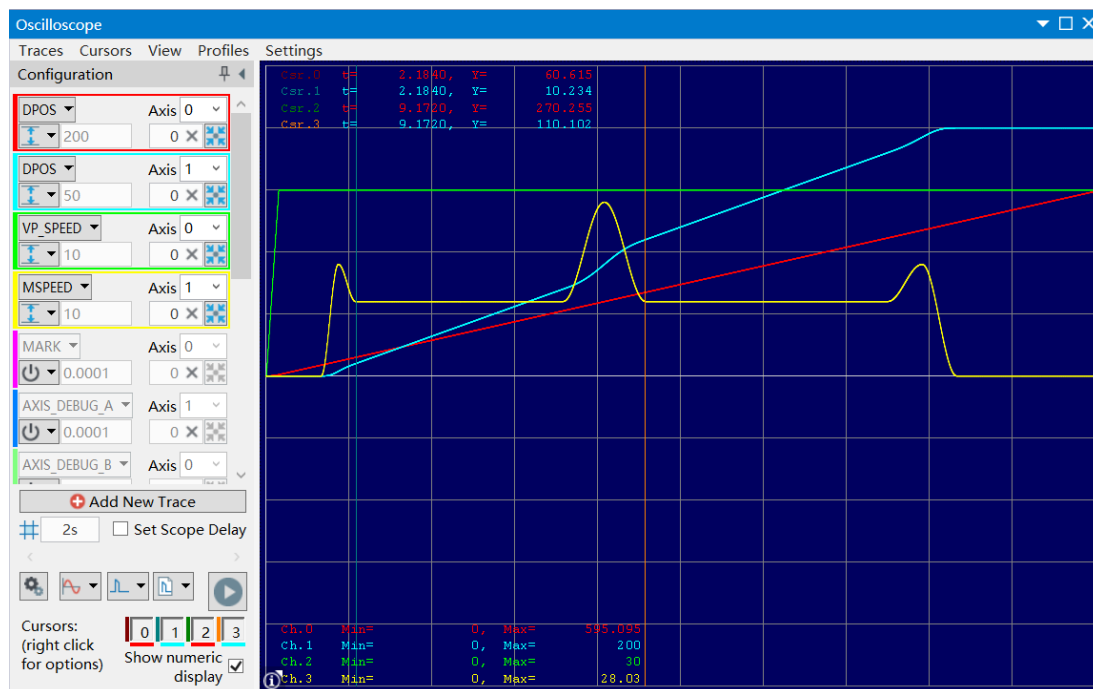
base_sync_2 = 70 ' incremental distance of the base axis in the sync phase
 link_axisnum = 0 ' link axis number
 link_options.5 = TRUE ' merge 2 consecutive ROTARYLINK commands
 link_options.2 = TRUE ' Sine profile
 sync_pos_1 = 60 ' Sync position of first ROTARYLINK command
 sync_pos_2 = 270 ' Sync position of second ROTARYLINK command

BASE(1) ' base axis - 1 , link axis - 0

ROTARYLINK(base_dist, link_dist, base_acc, base_sync_1, link_axisnum, _
link_options, sync_pos_1)

MOVE(600) AXIS(0)

ROTARYLINK(base_dist, link_dist, base_acc, base_sync_2, link_axisnum, _
link_options, sync_pos_2)



Example 5:

Use Mark to decide where the ROTARYLINK should start. In this case, when the MARK of link axis changes to -1, the base axis starts the ROTARYLINK movement.

```

FOR i = 0 TO 1
  BASE(i)
  UNITS = 2^23/360
  SPEED = 100
  ACCEL = 50
  DECEL = 50
  JERK = 50

```



```
SERVO = ON
FE_LIMIT = 99999
DRIVE_FE_LIMIT = 99999
FE_RANGE = 5000
FS_LIMIT = 99999
RS_LIMIT = -99999
DEFPOS(0)
NEXT i
```

```
WDOG = ON
```

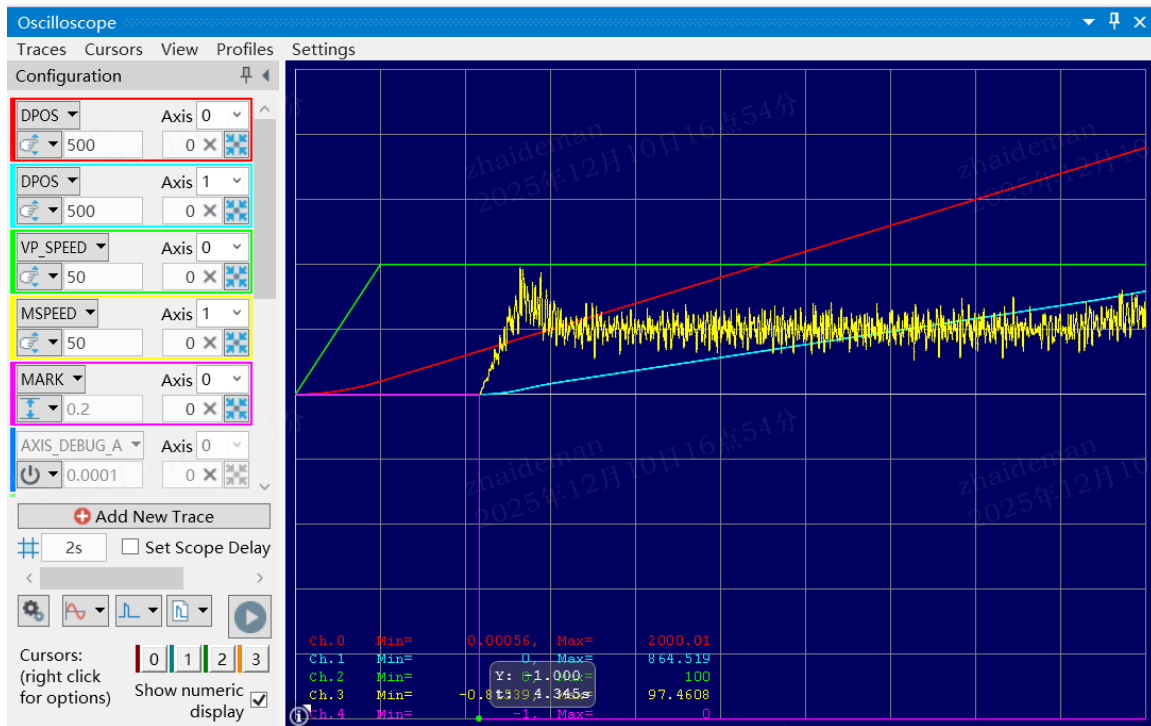
```
TRIGGER
```

```
BASE(0)
REGIST(0)
REGIST(20, 0, 1, 0, 0)
```

```
base_dist = 1000 ' incremental distance of base axis
link_dist = 2000 ' incremental distance of link axis
```

```
base_acc = 100
base_sync = 600
link_axisnum = 0
link_options = 1 ' Link commences exactly when registration event MARK occurs on link
axis
```

```
BASE(1)
ROTARYLINK(base_dist, link_dist, base_acc, base_sync, link_axisnum, link_options)
MOVE(2000) AXIS(0)
```



Example 6:

Repeat ROTALINK movements, keep the same accelerate distance and sync distance of slave axis. The start position of next movement is always smaller than the end position of current movement, so the movements were continuously merged.

master_axis = 0

link_axis_ = 1

BASE(link_axis_)

DEFPOS(0)

LIMIT_BUFFERED = 10

UNITS = 100000

SERVO=1

SRAMP = 0

ACCEL = 360

DECEL = 360

SPEED = 360

REP_OPTION = 1

REP_DIST=360

FE_LIMIT = 40

FE_RANGE = FE_LIMIT/2

AXIS_ENABLE=1

VP_MODE = 6

```
BASE(master_axis)
DEFPOS(0)
UNITS = 100000
ACCEL = 360
DECEL = 360
SPEED = 360
FE_LIMIT = 10
FE_RANGE = FE_LIMIT/2
REP_DIST=360000000
REP_OPTION = 1
JERK = 1000
VP_MODE = 6
DEFPOS(0)
```

```
WA(10)
WDOG = 1
WA(1000)
```

```
WA(50)
moveoptions = 0
moveoptions.0 = FALSE ' Start
moveoptions.1 = FALSE ' Start
moveoptions.2 = TRUE ' Profile
moveoptions.3 = FALSE ' Profile
moveoptions.4 = FALSE ' Profile
moveoptions.5 = TRUE ' Merge
```

```
distance = 360
linkdist = 360
accdist = 50
syncdist = 100
```

```
pos_off = 50
```

```
repeat_dist = 330
```

```
DEFPOS(0)
```

```
BASE(link_axis_)
MERGE = 0
start_pos = pos_off
PRINT #0, start_pos
FORWARD AXIS(master_axis)
```

WHILE (1)

TRIGGER

ROTARYLINK (distance, linkdist, accdist, syncdist, master_axis, moveoptions,start_pos)

start_pos = start_pos + repeat_dist

ROTARYLINK (distance, linkdist, accdist, syncdist, master_axis, moveoptions,start_pos)

start_pos = start_pos + repeat_dist

ROTARYLINK (distance, linkdist, accdist, syncdist, master_axis, moveoptions,start_pos)

start_pos = start_pos + repeat_dist

ROTARYLINK (distance, linkdist, accdist, syncdist, master_axis, moveoptions,start_pos)

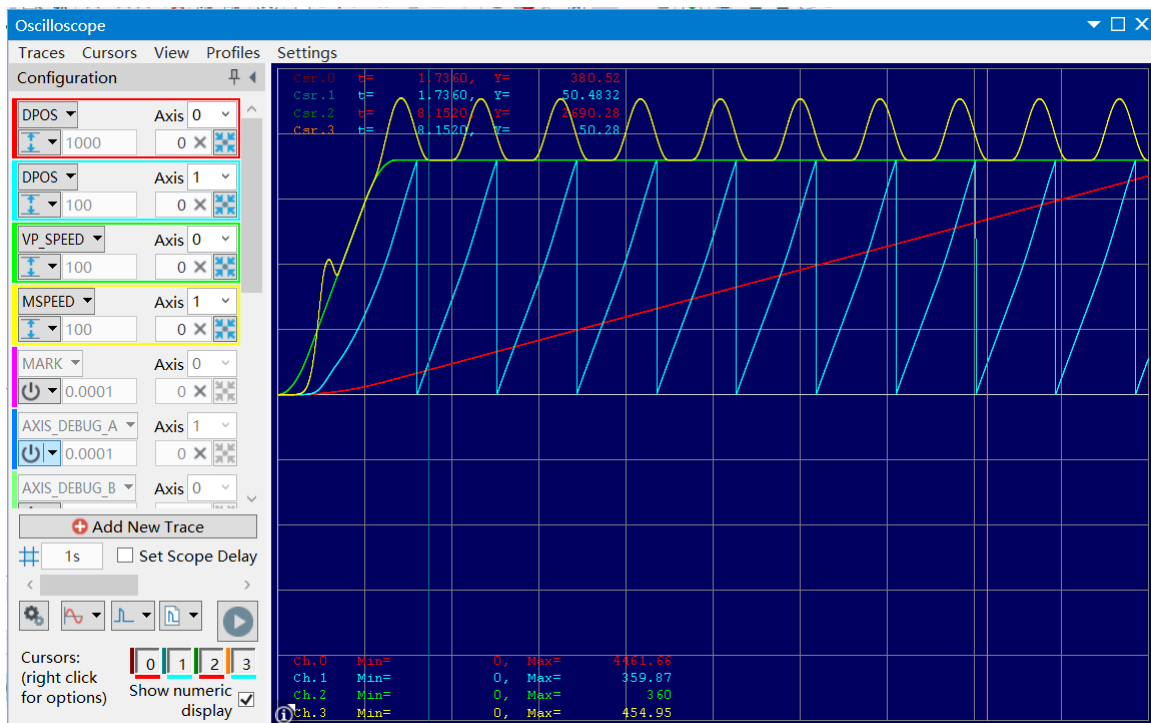
start_pos = start_pos + repeat_dist

WA(10)

WAIT UNTIL MOVES_BUFFERED < 2

'WAIT IDLE AXIS(10)

WEND



See Also:

MOVELINK, SYNC, FLEXLINK, CAMBOX